

CS5344 AY24-25 Sem2 Course Project Proposal for Group 2

Title: Search engine for travel activity ideas based on online content

Goal:

To build an engine that, from the input of the distance one is willing to travel from Singapore, present a list of activities with thumbnail pictures of the form
<list of similar activity keywords> <countries one can do it> <thumbnail images> <links to references>

Based on free content posted by online influencers.

Due to time constraints, we would only demonstrate the scraping of one or two websites as a proof of concept, and assume that it could be extended in the future for scraping more sites by simple, possibly A.I. powered customisation for each new site's html document structure and site navigation. We may also confine the scope to nearby countries for now.

Data Sources:

- Scraping the content of only one or two of the websites as a proof of concept:
 - <https://travelerfolio.com/>
 - <https://thesmartlocal.com/>
 - <https://alvinology.com/>
 - <https://iwandered.net/>
- <https://simplemaps.com/data/world-cities> to match location keywords to physical coordinates and for a corpus of location keywords
- <https://www.npmjs.com/package/verb-corpus> to find verbs in text to make sense of what people do with regard to destinations written about

Data Preprocessing:

- Cleaning and tokenizing the scraped content
- Dividing the contents of single pages that really are listicles. Header text would contextualise the content beneath it until the next header.
We can call a single activity description an *Activity Snippet*
- Normalise and resize the scraped images as thumbnails for presenting the Travel Activity ideas in the engine's interface

Data Analysis:

- Find meaningful phrases from scraped text using frequent items and extracting location phrases (e.g. JB, Singapore, khao kho) and verbs
- Find similar *Activity Snippets* across sources (different blogs may mention the same activity, or a similar thing can be done in different countries)
- Try to make sense of what the action is in an *Activity Snippet* (Eat? Sightsee? Museum?)
- Tag *Activity Snippets* by location
- Tag scraped images by location, activities, meaningful phrases

Challenges:

- Determining the action in an Activity Snippet is difficult because actions can be described differently, for example one could say 'take lunch' instead of 'eat'. We may have to supply some helpful related words by ourselves, or find such a data source